Math 22 - 100 Integrals

August 27, 2010

1. $\displaystyle\int x \sin^{-1}(x)\, dx$
2. $\displaystyle\int_0^1 \ln(1+x)\, dx$
3. $\displaystyle\int \sec^3(3x)\, dx$
4. $\displaystyle\int_0^1 e^{\sqrt{x}}\, dx$
5. $\displaystyle\int \cos^3(2x)\sin^2(2x)\, dx$
6. $\displaystyle\int \cos^4(x)\, dx$
7. $\displaystyle\int \tan(x)\sec^5(x)\, dx$
8. $\displaystyle\int \tan(x)\sec^6(x)\, dx$
9. $\displaystyle\int \frac{1}{(x^2+16)^{\frac{3}{2}}}\, dx$
10. $\displaystyle\int \frac{1}{x^2\sqrt{9-x^2}}\, dx$
11. $\displaystyle\int \frac{\sqrt{9-x^2}}{x}\, dx$
12. $\displaystyle\int \frac{x}{(x^2+1)^2}\, dx$
13. $\displaystyle\int \frac{x^3+2}{x(x-2)^3}\, dx$
14. $\displaystyle\int \frac{2}{x+x^3}\, dx$
15. $\displaystyle\int \frac{x^3-20x^2-63x-198}{x^4-81}\, dx$
16. $\displaystyle\int \frac{x-1}{(x+2)^5}\, dx$
17. $\displaystyle\int \frac{x}{\sqrt{4+4x-x^2}}\, dx$
18. $\displaystyle\int \frac{x}{x^2+4x+5}\, dx$
19. $\displaystyle\int \frac{\sqrt[3]{x+27}}{x}\, dx$
20. $\displaystyle\int \frac{\sin(x)}{3+2\cos(x)}\, dx$
21. $\displaystyle\int e^{2x}\sin(3x)\, dx$
22. $\displaystyle\int \cos(\ln(x))\, dx$
23. $\displaystyle\int \sin^3(x)\cos^3(x)\, dx$
24. $\displaystyle\int \cot^2(3x)\, dx$

25. $\displaystyle\int \frac{x}{\sqrt{8-x^2}}\,dx$

26. $\displaystyle\int \frac{1}{x\sqrt{4x^2-9}}\,dx$

27. $\displaystyle\int \frac{x^5-x^3+1}{x^3+2x^2}\,dx$

28. $\displaystyle\int \frac{x^3}{x^3-3x^2+9x-27}\,dx$

29. $\displaystyle\int \frac{1}{x^{3/2}+x^{1/2}}\,dx$

30. $\displaystyle\int \frac{2x+1}{(x+5)^{100}}\,dx$

31. $\displaystyle\int \frac{1}{\sqrt{7+5x^2}}\,dx$

32. $\displaystyle\int \frac{2x+3}{x^2+9}\,dx$

33. $\displaystyle\int \cot^6(x)\,dx$

34. $\displaystyle\int \cot^5(x)\csc(x)\,dx$

35. $\displaystyle\int x^3\sqrt{x^2-36}\,dx$

36. $\displaystyle\int (\sin x)10^{\cos(x)}\,dx$

37. $\displaystyle\int e^x\sec(e^x)\,dx$

38. $\displaystyle\int x\tan(x^2)\,dx$

39. $\displaystyle\int x^2\sin(7x)\,dx$

40. $\displaystyle\int \sin(2x)\cos(x)\,dx$

41. $\displaystyle\int \sin^3(x)\cos^{1/2}(x)\,dx$

42. $\displaystyle\int \sin(3x)\cot(3x)\,dx$

43. $\displaystyle\int e^x\sqrt{1+e^x}\,dx$

44. $\displaystyle\int x(9x^2+25)^{-1/2}\,dx$

45. $\displaystyle\int x^2(9x^2+25)^{-1/2}\,dx$

46. $\displaystyle\int \frac{3x+2}{x^2+8x+25}\,dx$

47. $\displaystyle\int \sec^2(x)\tan^2(x)\,dx$

48. $\displaystyle\int \sin^2(x)\cos^5(x)\,dx$

49. $\displaystyle\int x\cot(x)\csc(x)\,dx$

50. $\displaystyle\int (1+\csc(2x))^2\,dx$

51. $\displaystyle\int x^2(27-x^3)^{1/3}\,dx$

52. $\displaystyle\int x(\ln x)^2\,dx$

53. $\displaystyle\int \sqrt{x}\sin\sqrt{x}\,dx$

54. $\displaystyle\int x\sqrt{3-5x}\,dx$

55. $\displaystyle\int \frac{e^{3x}}{1+e^{x}}\,dx$

56. $\displaystyle\int \frac{e^{2x}}{4+e^{4x}}\,dx$

57. $\displaystyle\int \left[x^{2}-\operatorname{sech}^{2}(4x)\right]\,dx$

58. $\displaystyle\int x\cosh(x)\,dx$

59. $\displaystyle\int x^{2}e^{-4x}\,dx$

60. $\displaystyle\int x^{5}\sqrt{x^{3}+1}\,dx$

61. $\displaystyle\int \frac{3}{\sqrt{11-10x-x^{2}}}\,dx$

62. $\displaystyle\int \frac{12x^{3}+7x}{x^{4}}\,dx$

63. $\displaystyle\int \tan(7x)\cos(7x)\,dx$

64. $\displaystyle\int e^{1+\ln(5x)}\,dx$

65. $\displaystyle\int \frac{4x^{2}-12x-10}{(x-2)(x^{2}-4x+3)}\,dx$

66. $\displaystyle\int \frac{1}{x^{4}\sqrt{16-x^{2}}}\,dx$

67. $\displaystyle\int (x^{3}+1)\cos(x)\,dx$

68. $\displaystyle\int (x-4)^{2}(x-1)\,dx$

69. $\displaystyle\int \frac{\sqrt{4-9x^{2}}}{x^{2}}\,dx$

70. $\displaystyle\int \frac{4x^{3}-15x^{2}-6x+81}{x^{4}-18x^{2}+81}\,dx$

71. $\displaystyle\int \frac{x^{2}-4x+3}{\sqrt{x}}\,dx$

72. $\displaystyle\int \frac{\cos^{3}(x)}{\sqrt{1+\sin(x)}}\,dx$

73. $\displaystyle\int \frac{x^{3}}{\sqrt{25-x^{2}}}\,dx$

74. $\displaystyle\int \frac{x}{9-4x^{2}}\,dx$

75. $\displaystyle\int \frac{1-2x}{x^{2}+12x+35}\,dx$

76. $\displaystyle\int \frac{75}{x^{2}-6x+18}\,dx$

77. $\displaystyle\int \tan^{-1}(5x)\,dx$

78. $\displaystyle\int \sin^{4}(3x)\,dx$

79. $\displaystyle\int \frac{e^{\tan(x)}}{\cos^{2}(x)}\,dx$

80. $\displaystyle\int \frac{x}{\csc(5x^{2})}\,dx$

81. $\displaystyle\int [7-\cot(3x)]^{2}\,dx$

82. $\displaystyle\int x(x^{2}+5)^{3/4}\,dx$

83. $\displaystyle\int \frac{1}{x(\sqrt{x}+\sqrt[4]{x})}\,dx$

84. $\displaystyle\int \frac{x}{\cos^{2}(3x)}\,dx$

85. $\displaystyle\int \frac{\sin(x)}{\sqrt{1+\cos(x)}}\,dx$

86. $\displaystyle\int \frac{4x^2-6x+4}{(x^2+4)(x-2)}\,dx$

87. $\displaystyle\int \frac{x^2}{(16+x^2)^2}\,dx$

88. $\displaystyle\int \sin^3(x)\cos^4(x)\,dx$

89. $\displaystyle\int \tan^3(x)\sec(x)\,dx$

90. $\displaystyle\int \frac{x}{\sqrt{9+4x^2}}\,dx$

91. $\displaystyle\int \frac{2x^3+4x^2+10x+13}{x^4+9x^2+20}\,dx$

92. $\displaystyle\int \frac{\sin x}{(1+\cos x)^3}\,dx$

93. $\displaystyle\int \frac{(x^2-3)^2}{x}\,dx$

94. $\displaystyle\int \cot^2(x)\csc(x)\,dx$

95. $\displaystyle\int x^{3/2}\ln(x)\,dx$

96. $\displaystyle\int \frac{x}{\sqrt[3]{x}-1}\,dx$

97. $\displaystyle\int \frac{x^2}{\sqrt[3]{2x+3}}\,dx$

98. $\displaystyle\int \frac{1-\sin(x)}{\cot(x)}\,dx$

99. $\displaystyle\int x^3 e^{(x^2)}\,dx$

100. $\displaystyle\int (x+2)^2(x+1)^{10}\,dx$